Aekank Patel

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SUMMARY

Data Science Professional with experience across 3 roles and 19+ projects in Python, SQL, and machine learning, building predictive models and LLM-powered systems across fraud detection, healthcare analytics, and financial document intelligence. Skilled in classification, deep learning, and retrieval-augmented generation. Experienced in cross-functional collaboration and end-to-end model development.

EDUCATION

Stevens Institute of Technology, Master of Science (MS), Data Science, Hoboken, NJ, **GPA: 3.97/4.0** **Sep 2024 – May 2026**

Manipal Institute of Technology, Bachelor of Technology, Mechatronics Engineering, Manipal, KA, **GPA: 3.2/4.0** **Oct 2020 – Jul 2024**

WORK EXPERIENCE

Graduate Teaching Assistant | **Stevens Institute of Technology** | Hoboken, NJ **Sep 2025 – May 2026**

- Standardized rubric-aligned evaluation across 1,000+ submissions in three graduate data science courses spanning linear algebra, foundational mathematics, and probability theory, by managing grading and feedback workflows in Canvas LMS and Excel.
- Maintained accurate grade records for 89 students under tight deadlines, by cross-checking submission logs against gradebook entries.

Research and Development Intern | **Matrix ComSec Pvt. Ltd.** | Vadodara, GJ **Jan 2024 – May 2024**

- Prepared 100,000+ annotated images across pose and crowd-density segments for model training, correcting 12% class imbalance and filtering mislabeled or outlier frames, by building automated data profiling and quality-filtering pipelines.
- Diagnosed an 8% model performance gap across pose, crowd-density, and camera-angle cohorts, by conducting systematic error analysis that directly informed architecture refinement before deployment.
- Deployed a real-time fall detection system to live camera feeds, reaching 91.39% test accuracy across multi-person video streams, by optimizing the inference pipeline and integrating a hybrid CNN-MediaPipe and YOLOv5 architecture for production use.
- Benchmarked 4 deep learning architectures (CNN, RNN, BodyPix, R-CNN) for fall detection, selecting a top-performing architecture with a 6% accuracy improvement over baseline approaches, through a literature review and comparative evaluation.

Data Science Intern | **Nimit Electronics and Equipments** | Vadodara, GJ **May 2023 – Dec 2023**

- Structured 80,000+ surveillance, access-control, and IoT sensor records into model-ready feature sets for downstream classification and anomaly-detection models, by querying and consolidating the data with SQL.
- Built a standardized feature pipeline across 10+ heterogeneous datasets, by developing Python-based ETL scripts to extract, clean, transform, and validate records ahead of downstream modeling.
- Engineered features capturing device health and access-activity patterns across 5+ device types, multiple locations, and time periods, by analyzing distribution, correlation, and outlier patterns in Python, Pandas, and NumPy.
- Achieved 94.83% accuracy in incident classification and anomaly detection, by training and evaluating multiple machine learning models in Scikit-learn and selecting the best-performing approach based on precision, recall, and F1-score.

TECHNICAL SKILLS

Programming & Data Analysis: Python (Pandas, NumPy, Scikit-learn), SQL (Joins, CTEs, Aggregations, Window Functions), EDA, Data Cleaning, ETL, Feature Engineering

Machine Learning & Deep Learning: Anomaly Detection, Random Forest, XGBoost, Class Imbalance Handling, CNN, RNN, TensorFlow

Deployment & Tools: Retrieval-Augmented Generation (RAG), LlamaIndex, LLaMA, Groq, Flask REST APIs, Tableau, Streamlit, Git, AWS

PROJECTS

Patient No-Show Risk Dashboard – Tableau, GitHub **Mar 2026 – Apr 2026**

- Reached 0.966 AUC predicting patient no-shows across 110,000+ appointments using a Random Forest model, by engineering behavioral and scheduling features in a structured SQLite database to project reductions in preventable no-show revenue loss.
- Uncovered a misleading SMS-reminder correlation (28% vs 17% no-show rate) driven by targeting bias, by analyzing appointment patterns across weekday, clinic, age group, and lead-time dimensions.

FinRAG: Financial Document Intelligence System– Demo, GitHub **Jun 2025 – Aug 2025**

- Developed a RAG system answering natural-language questions over 25 SEC filings and earnings transcripts, achieving an 80% evaluation pass rate across 10 test queries, by chunking documents into 14,055 indexed nodes with LlamaIndex.
- Redesigned the inference pipeline for cloud deployment after local Ollama proved incompatible with Streamlit Cloud, by bypassing LlamaIndex's LLM layer to call the Groq SDK directly for generation.

FraudGen: Real-Time Fraud Detection and Risk Explanations – Demo, GitHub **Mar 2025 – May 2025**

- Classified financial transactions into fraud risk tiers, achieving 96.55% recall and 0.9995 ROC-AUC across 6.36M transactions, by engineering behavioral, transactional, and risk-based features for an XGBoost classifier.
- Reduced manual fraud-review workload by generating plain-English risk explanations for high-severity transactions, using a LLaMA-powered pipeline served through a Flask REST API with SQLite storage.

CERTIFICATES

AWS Data Engineering, AWS Cloud Foundations, Google Data Analytics, IBM AI Engineering, Bloomberg Market Concepts